

Technical Datasheet - VBF-T8R2-DSH-01 (t8_r2, final design V_G)

	Value
	Nominal 5.000 (parameter set); simulated 1C DFN 5.0351 (cell/r3_VG_1c_dfn.json:capacity_ah)
	2.5 - 4.2 (parameter set Lower/Upper voltage cut-off); midpoint voltage 3.9029 (cell/r3_VG_energy.json:midpoint_voltage_v)
	18.229 (discharge energy, time integral of V*I_1C; cell/r3_VG_energy.json:energy_wh)
	483.4 (cell/r3_VG_energy.json:energy_density_wh_kg, contract formula: energy_wh / mass_kg; mass = sum layer thickness x (1-porosity) x density x area)
	935.3 (cell/r3_VG_energy.json:energy_density_wh_l; volume = stack thickness x area)
rate	5C verified: capacity 4.8492 Ah, retention 0.9631 vs 1C (contract target >= 0.90 at 5C; DFN, cell/r3_VG_5c_dfn.json / cell/r3_VG_1c_dfn.json)
	4C charge at 45 C: T_max 351.93 K (78.8 C, +33.78 K over ambient); no plating (anode potential min 0.0215 V > 0; cell/r3_VG_safety.json, protocol 4C_charge_45C, therm)
	Simulated conditions only: discharge at default ambient (reference temp 298.15 K, T_max 306.15 K at 1C); 4C fast charge at 45 C ambient (protocol). Full envelope not cha
	Not simulated (aging model not exercised; task contract has no cycle-life criterion) - not fabricated
	2.7305 mOhm, derived by calc-energy as (V(t=0) - V(t=10%))/I_1C from the 1C curve (cell/r3_VG_energy.json:dcr_ohm). Pulse-based DCR: N/A (requires physical experim
	4C/45C charge: no plating (PASS vs contract plated=false); T_max record 351.93 K (no contract threshold)
	stack thickness 150.6 um; electrode height 65.0 mm x width 1990.8 mm; mass 37.71 g (contract caliber, electrolyte excluded), annotated electrolyte 6.40 g